



Step-by-Step: Different data perspectives

This reading outlines the steps the instructor performs in the next video, [Different data perspectives](#). The video teaches you different methods data analysts use to view data differently and how looking at different views leads to more efficient and effective data cleaning.

Keep this step-by-step guide open as you watch the video. It can serve as a helpful reference if you need additional context or clarification while following the video steps. This is not a graded activity, but you can complete these steps to practice the skills demonstrated in the video.

What you'll need

If you'd like to follow along with the examples in this video, choose a spreadsheet tool. Google Sheets or Excel are recommended.

To access the spreadsheet the instructor uses in this video, click the link to the template to create a copy of the dataset. If you don't have a Google account, download the data directly from the attachments below.

Link to template: [Cosmetics, Inc.](#)

OR

Example 1: Pivot tables

A pivot table is a data summarization tool. It can be used in data processing and in data cleaning, for which pivot tables offer a quick, clutter-free view of your data. Pivot tables help sort, reorganize, group, count, total, or average data in a dataset.

1. In the Cosmetics Inc. spreadsheet, select the data you'll include. In this case, select all of the data in Sheet 1 of the spreadsheet by selecting cell A1 then dragging your cursor to cell F31.
2. Select Insert, then Pivot Table. Choose New sheet and Create. Google Sheets creates a new sheet where you can define the pivot table.
3. Use the Pivot table editor to add specific data to your pivot table.
 - a. In the Pivot table editor panel, next to Rows, select Add.
 - b. From the columns list, select Total.
 - c. Below Rows, from the Order dropdown list, select Descending to put the most profitable items at the top.
 - d. Next to Rows, select Add.
 - e. From the column list, select Products.
 - f. Notice that the top two most ordered products are 15143Exfo and 32729Masc. The rest of the orders total less than \$10,000.

Example 2: **VLOOKUP**

VLOOKUP is a spreadsheet function that vertically searches for a certain value in a column to return a corresponding piece of information. It's rare for all of the data an analyst will need to be in the same place. Usually, you'll have to search across multiple sheets or even different databases. **VLOOKUP** helps bring the information together.

In the previous example, you found the product codes of the most ordered products. Now, you'll use **VLOOKUP** to find the names of these products.

1. Select the Sheet 1 tab to navigate to Sheet 1 of the spreadsheet.
2. Select cell H2.
3. Enter **=VLOOKUP(A2, 'Sheet 2'!A1:B31, 2, false)**
 - a. Note: This references information in another sheet. Make sure you have Sheet 2 in your workbook.
 - b. This formula will take the value in cell A2 of Sheet 1 and check for that value in Sheet 2 among the cells from A1:B31 in the 2nd column (which corresponds with the 2 in the formula). Because the formula includes "**false**," it will search only for an exact match. It will then output the value of column B in Sheet 2 as the result.
4. Press Enter to input the formula. The result is LashX Mascara.
5. Next, select the cell and drag the fill handle in the lower-right corner down to populate the other cells in the sheet with the formula.
6. To find the names of the two most profitable products you identified in the previous example use the find and replace tool.
 - a. Select Edit > Find and Replace.
 - b. In the Find text box, enter the product code for the most profitable product, 15143Exfo.
 - c. Select This sheet from the dropdown list next to Search.
 - d. Select Find to find any cells in this sheet that contain this product code.
 - e. Notice that cell A31 is a match. This means the VLOOKUP search you ran in Column H31 contains the name of the most profitable product: SoSoft Exfoliator.
 - f. Repeat steps a-d with the product code 32729Masc to find the product name of the second most profitable product. Cell A27 contains 32729Masc, so the product name is Darkest Lashes Mascara.

Example 3: Plotting

The plotting tool allows analysts to quickly create a graph, chart, table, or other visual from data. Plotting is useful for identifying skewed data or outliers.

1. In Sheet 1 of the *Cosmetics, Inc.* spreadsheet, select column B, which contains the prices.
2. Select Insert > Chart.
 - a. If the chart created is not a column chart, select Column chart from the dropdown menu under Chart type in the Chart editor.
 - b. Select and drag the chart to the right so you can view the data in the sheet.
3. Check for obvious outliers and fix them in the spreadsheet. For example, you might notice that an item in the middle of the chart has an extremely low price of \$0.73. The decimal point is in the wrong place. In cell B14, correct this price to \$7.30, and notice that Google Sheets automatically updates the chart.

Key Concepts:

- **VLOOKUP:** VLOOKUP is a spreadsheet function that searches for a value in a column and returns a corresponding piece of information from another column.
- **Pivot Table:** A pivot table is a data summarization tool that allows users to sort, reorganize, group, count, total, or average data in a dataset.
- **Plotting:** Plotting tools enable analysts to create visual representations of data, helping to identify trends, outliers, and patterns.